BMC Women's Health

RESEARCH

Open Access

Association of PAX1/JAM3 gene methylation, HR-HPV, and TCT with high-grade cervical lesions in a high-risk cohort: a multinomial logistic regression analysis

Tongyao Geng[1†], Sijia Li[2†], Yanan Bian[3], Xiaodong Li[1*] and Yahui Bian[1*]

Abstract

Objective To investigate the risk factors associated with pathological progression of cervical lesions, specifically focusing on high-risk human papillomavirus (HR-HPV) infection, ThinPrep Cytologic Test (TCT), and PAX1/JAM3 gene methylation in a high-risk colposcopy-referred cohort.

Methods A total of 409 patients who underwent colposcopy-directed biopsy at the First Hospital of Hebei Medical University between January, 2023 and April, 2025 due to abnormal TCT results, HR-HPV infection, or positive PAX1/JAM3 gene methylation were analyzed. The sensitivities of different methods were compared, and unordered multivariate logistic regression was used to analyze the risk factors for high-grade cervical lesions and pathological progression.

Results Univariate analysis demonstrated that age, gravidity, age at first sexual intercourse, number of sexual partners, menopausal status, HR-HPV infection, TCT results, and PAX1/JAM3 methylation were significantly associated with pathological grade (all $P < 0.05$). In the multinomial regression analysis, compared to the inflammation group, both other high-risk HPV (OR = 3.930) and very-high-risk (16/18) HPV infections (OR = 3.725), as well as non-NILM TCT (OR = 2.435), were significantly associated with an increased risk of LSIL. For the progression to HSIL, very-high-risk HPV (OR = 6.801), non-NILM TCT (OR = 2.399), positive PAX1 methylation (OR = 4.145), and positive JAM3 methylation (OR = 8.215) were identified as strong independent risk factors.

Conclusion In a high-risk colposcopy-referred cohort, HR-HPV infection, non-NILM TCT results, and positive PAX1/JAM3 gene methylation are independently associated with the presence of high-grade cervical lesions. These biomarkers have the potential to enhance clinical risk stratification and inform decision-making in cervical cancer screening. However, their clinical utility and cost-effectiveness need to be further evaluated through larger, prospective studies to validate their role in routine clinical practice.

†Tongyao Geng and Sijia Li contributed equally to this work.

*Correspondence:
Xiaodong Li
lixiaodong@hebmu.edu.cn
Yahui Bian
1002398763@qq.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Keywords Cervical lesions, Risk factors, Multinomial logistic regression, PAX1/JAM3 methylation, Human papillomavirus

Cervical cancer is a common malignant tumor of the female reproductive system and poses a significant threat to women's health worldwide. In China, there were 132,800 new cases of cervical cancer and 49,800 deaths in 2022 alone [1]. The vast majority of cervical cancers are caused by persistent infection with human papillomavirus (HPV), making it one of the few cancers with a well-defined etiology that is also preventable [2].The combination of etiology-based HR-HPV testing and ThinPrep Cytologic Test (TCT) cytological assessment has laid the foundation for cervical cancer screening [3]. However, the TCT requires specialized equipment and well-trained personnel, which hinders its widespread implementation in resource-limited regions [4]. Although HR-HPV testing offers high sensitivity, the vast majority of infections are transient and resolve spontaneously. This leads to an increase in referrals for colposcopic evaluation, highlighting the need for additional risk stratification tools [5, 6]. Epigenetic alterations are critical for cancer development. Paired box gene 1 (PAX1) and Junctional Adhesion Molecule 3 (JAM3) are important genes where aberrant methylation is closely associated with tumor progression and cervical lesions [7–10]. This study aims to analyze the risk factors for developing cervical lesions in a high-risk hospital-based cohort using a multinomial logistic regression model, exploring the association of PAX1/JAM3 methylation alongside traditional HPV and TCT metrics.

Materials and methods

Study population

This study involves a high-risk, hospital-based cohort. Patients who underwent colposcopy-directed biopsy at the First Hospital of Hebei Medical University between January 2023 and April 2025 due to abnormal TCT results, HR-HPV infection, or positive PAX1/JAM3 gene methylation were retrospectively selected. It should be noted that as this population consists of individuals already referred due to screening abnormalities, it represents a high-risk cohort rather than a general screening population.

Inclusion and exclusion criteria

Inclusion criteria: (1) Sexually active females aged ≥ 18 years. (2) No use of vaginal medication for 3 days prior to the examination. (3) Participants gave written informed consent in accordance with the Declaration of Helsinki guidelines.

Exclusion criteria: (1) Presence of contraindications to TCT, HR-HPV testing, PAX1/JAM3 gene testing, or colposcopy. (2) Presence of severe organic diseases.

Ultimately, a total of 409 patients were enrolled in this study. The histopathological results from the colposcopy-directed biopsies were as follows. Normal or cervicitis: 168 cases (41.1%), low-grade squamous intraepithelial lesion (LSIL): 93 cases (22.7%), and high-grade squamous intraepithelial lesion (HSIL):148 cases (36.2%).

The study was approved by the Ethics Committee of The First Hospital of Hebei Medical University. Initial approval was granted for a pilot cohort of 100 patients. To ensure sufficient statistical power, a subsequent application to expand the sample size to 409 patients was submitted and approved by the committee.

Observation indicators

TCT detection

A sampling brush was inserted 1 cm into the cervical canal, rotated five times in a single direction, and then placed into specimen preservation solution. According to the criteria of The Bethesda System (TBS), cytological diagnoses were classified as: negative for intraepithelial lesion or malignancy (NILM), atypical squamous cells of undetermined significance (ASC-US), atypical squamous cells (ASC-H), cannot exclude HSIL, LSIL, and HSIL. For analysis, the cases were categorized into an NILM group ($n = 228$) and a Non-NILM group ($n = 181$). Given that the TCT results in this study included NILM, ASC-US, and LSIL, with an insufficient number of LSIL cases, the ASC-US and LSIL groups were combined into a non NILM group.

HR-HPV detection

HPV genotyping was performed using cervical samples collected with a Brush, which was rotated five times at the cervical os to ensure adequate cell collection. The brush head was then placed into cell preservation solution and transported to the laboratory. DNA was extracted using the Qiagen DNA Extraction Kit according to the manufacturer's instructions. DNA concentration was measured using a Nanodrop spectrophotometer to ensure the sample was suitable for subsequent PCR analysis. HPV genotyping was performed using a commercially available PCR-based HPV genotyping kit, following the manufacturer's standard protocol. PCR amplification products were analyzed by agarose gel electrophoresis to detect HPV types. Each PCR run included negative and positive controls to ensure the reliability of the results, and all positive results were repeated for confirmation. Results were reported as positive or negative for each HPV type based on the PCR amplification products.

Table 1 Baseline characteristics of study participants ($n=409$)

Variable	Total ($n=409$)	Inflammation ($n=168$)	LSIL ($n=93$)	HSIL ($n=148$)	statistic	*P*
Age, Median (Q1, Q3)	39.0 (30.00, 50.00)	40.0 (30.75, 52.00)	43.0 (34.00, 51.00)	36.0 (29.75, 46.00)	8.515	0.014
Marital status					2.550	0.279
Unmarried	57 (13.94)	20 (11.90)	11 (11.83)	26 (17.57)		
Married	352 (86.06)	148 (88.10)	82 (88.17)	122 (82.43)		
Gravidity, n (%)					9.533	0.049
0	71 (17.36)	32 (19.05)	14 (15.05)	25 (16.89)		
1–2	171 (41.81)	80 (47.62)	30 (32.26)	61 (41.22)		
≥ 3	167 (40.83)	56 (33.33)	49 (52.69)	62 (41.89)		
Parity, n (%)					5.047	0.283
0	98 (23.96)	41 (24.40)	17 (18.28)	40 (27.03)		
1–2	288 (70.42)	121 (72.02)	68 (73.12)	99 (66.89)		
≥ 3	23 (5.62)	6 (3.57)	8 (8.60)	9 (6.08)		
Mode of delivery, n (%)					5.503	0.239
No delivery	95 (23.23)	39 (23.21)	16 (17.20)	40 (27.03)		
Vaginal delivery	240 (58.68)	93 (55.36)	62 (66.67)	85 (57.43)		
Cesarean section	74 (18.09)	36 (21.43)	15 (16.13)	23 (15.54)		
Postcoital bleeding, n (%)					1.963	0.375
No	352 (86.06)	146 (86.90)	83 (89.25)	123 (83.11)		
Yes	57 (13.94)	22 (13.10)	10 (10.75)	25 (16.89)		
BMI, Median (Q1, Q3)	22.10 (20.13, 24.46)	22.36 (20.07, 24.62)	21.61 (20.43, 24.65)	22.07 (20.02, 24.19)	0.218	0.897
HPV vaccination, n (%)					0.009	0.996
No	281 (68.70)	115 (68.45)	64 (68.82)	102 (68.92)		
Yes	128 (31.30)	53 (31.55)	29 (31.18)	46 (31.08)		
Age at first sexual intercourse, Median (Q1, Q3)	22 (20, 24)	22 (20, 24)	22 (21, 24)	21 (20, 23)	6.008	0.050
Number of sexual partners, n (%)					8.920	0.012
0–1	241 (58.92)	107 (63.69)	61 (65.59)	73 (49.32)		
≥ 2	168 (41.08)	61 (36.31)	32 (34.41)	75 (50.68)		
Menopausal status, n (%)					8.499	0.014
No	319 (78.00)	122 (72.62)	70 (75.27)	127 (85.81)		
Yes	90 (22.00)	46 (27.38)	23 (24.73)	21 (14.19)		
Contraceptive method, n (%)					Fisher	0.332

Table 1 (continued)

Variable	Total (*n* = 409)	Inflammation (*n* = 168)	LSIL (*n* = 93)	HSIL (*n* = 148)	statistic	*P*
No contraception	118 (28.85)	42 (25.00)	32 (34.41)	44 (29.73)		
Condom	221 (54.03)	98 (58.33)	43 (46.24)	80 (54.05)		
IUD (Intrauterine device)	52 (12.71)	23 (13.69)	14 (15.05)	15 (10.14)		
Withdrawal	18 (4.40)	5 (2.98)	4 (4.30)	9 (6.08)		
Family history of cancer, n (%)					0.304	0.859
No	291 (71.15)	122 (72.62)	65 (69.89)	104 (70.27)		
Yes	118 (28.85)	46 (27.38)	28 (30.11)	44 (29.73)		
HPV status, n (%)					36.970	< 0.001
Low-risk or negative	31 (7.58)	22 (13.10)	4 (4.30)	5 (3.38)		
Other high-risk	188 (45.97)	87 (51.79)	53 (56.99)	48 (32.43)		
Very-high-risk (types 16, 18)	190 (46.45)	59 (35.12)	36 (38.71)	95 (64.19)		
TCT, n(%)					22.653	< 0.001
NILM	228 (55.75)	117 (69.64)	45 (48.39)	66 (44.59)		
Non-NILM	181 (44.25)	51 (30.36)	48 (51.61)	82 (55.41)		
P1, n(%)					50.154	< 0.001
Negative	350 (85.57)	163 (97.02)	84 (90.32)	103 (69.59)		
Positive	59 (14.43)	5 (2.98)	9 (9.68)	45 (30.41)		
J3, n(%)					63.976	< 0.001
Negative	337 (82.40)	162 (96.43)	82 (88.17)	93 (62.84)		
Positive	72 (17.60)	6 (3.57)	11 (11.83)	55 (37.16)		

Abbreviations: *LSIL* Low-Grade Squamous Intraepithelial Lesion, *HSIL* High-Grade Squamous Intraepithelial Lesion, *BMI* Body Mass Index, *IUD* Intrauterine Device, *HPV* Human Papillomavirus, *TCT* ThinPrep Cytologic Test, *NILM* Negative for Intraepithelial Lesion or Malignancy, *Non-NILM* Abnormal TCT result

Normally distributed quantitative data were presented as mean ± SD, while non-normally distributed data were presented as median and interquartile range (IQR). Categorical data were presented as frequency (percentage)

PAX1/JAM3 gene methylation detection

Exfoliated cervical cells were collected with a disposable cervical sampling swab and stored in a sample tube containing cell preservation solution. The purity and concentration of the extracted DNA were determined using a real-time fluorescence quantitative PCR system. A total of 200 ng of DNA from each sample underwent bisulfite conversion. This converted DNA was then analyzed according to the manufacturer's instructions for the Human PAX1 and JAM3 Gene Methylation Detection Kit (PCR-Fluorescence Probe Method; Beijing Genecast Biotechnology Co., Ltd.).The ΔCt values for the target genes were recorded, with ΔCtP representing the PAX1 result and ΔCtJ representing the JAM3 result.Methylation status was determined using the threshold ΔCt values recommended by the manufacturer: a sample was considered positive for PAX1 methylation if $\Delta CtP \leq 6.6$, and positive for JAM3 methylation if $\Delta CtJ \leq 10$. These thresholds were selected based on the manufacturer's guidelines and prior validation studies, which have demonstrated their effectiveness in distinguishing methylated from non-methylated samples. These thresholds are widely used in clinical research and diagnostic applications.

The Human PAX1 and JAM3 Gene Methylation Detection Kit has been validated in various clinical studies and has shown reliable results in similar patient populations, confirming its applicability in our study group [11].

To ensure the accuracy and reproducibility of our results, we strictly adhered to the quality control

measures outlined in the kit's instructions. Intra-assay variability was assessed by performing three replicate assays on a single sample, and the coefficient of variation (CV) for ΔCt values was found to be below 5%. Inter-assay variability was evaluated by conducting duplicate assays on different days, with the CV between runs consistently remaining below 10%. These measures ensure the reliability and consistency of the assay in detecting PAX1 and JAM3 gene methylation.

Pathological examination

Tissues obtained from colposcopy-directed biopsy were sent for pathological examination, and the results were classified as chronic cervicitis, LSIL, or HSIL.

Statistical analysis

Descriptive analysis was performed initially. Normally distributed quantitative data were expressed as mean ± standard deviation ($\bar{x}\pm$ SD), while non-normally distributed data were expressed as median and interquartile range [M (IQR)]. Categorical data were presented as frequencies and percentages [n (%)] and compared using the Chi-square test or Fisher's exact test. Differences among groups for non-parametric continuous variables were assessed using the Kruskal-Wallis H test.

To identify risk factors associated with cervical lesions, a multinomial logistic regression model was utilized. Although cervical lesions (Inflammation → LSIL → HSIL) progress ordinally, a preliminary Test of Parallel Lines (proportional odds assumption) yielded a statistically significant result ($P = 0.024$), indicating the assumption for ordinal regression was violated. Therefore, a multinomial logistic regression approach was deemed statistically appropriate to independently identify risk factors for LSIL and HSIL. Model diagnostics included the Likelihood Ratio Test and Pearson's/Deviance goodness-of-fit tests. Multicollinearity was checked using the Variance Inflation Factor (VIF). All analyses were performed using R (version 4.4.1) and SPSS 26.0. A two-sided $P < 0.05$ was considered significant.

Results

Baseline characteristics of the study population

A total of 409 participants were included in this study, with a median age of 39 years (interquartile range [IQR]: 30.00–50.00). Based on pathological diagnosis, the cohort was divided into three groups: Inflammation ($n = 168$, 41.1%), LSIL/CIN1 ($n = 93$, 22.7%), and HSIL/CIN2-3 ($n = 148$, 36.2%). The majority of participants were married (86.1%) and had 1–2 parity (41.8%). Regarding viral and cytological indicators, 46.0% of patients were infected with general high-risk HPV, 46.5% were infected with very-high-risk HPV, and 44.3% had high-risk TCT results. For the epigenetic markers, the overall positivity rates for PAX1 and JAM3 methylation were 14.4% (59/409) and 17.6% (72/409), respectively. Univariable analysis revealed statistically significant differences among the three pathological groups regarding age ($P = 0.014$), parity ($P = 0.049$), number of sexual partners ($P = 0.012$), menopausal status ($P = 0.014$), HPV infection status ($P < 0.001$), TCT results ($P < 0.001$), PAX1 methylation ($P < 0.001$), and JAM3 methylation ($P < 0.001$). The detailed baseline characteristics are summarized in Table 1.

Multinomial logistic regression model diagnostics

To identify independent risk factors for different grades of cervical lesions, variables with statistical significance in the univariable analysis were incorporated into a regression model. A preliminary Test of Parallel Lines indicated that the proportional odds assumption was violated ($P = 0.024$), precluding the use of ordinal logistic regression. Consequently, an unordered multinomial logistic regression model was employed. The Likelihood Ratio Test demonstrated that the final model was significantly superior to the null model ($\chi^2 = 145.916$, $P < 0.001$). Both the Pearson goodness-of-fit test ($\chi^2 = 761.944$, $P = 0.555$) and the Deviance goodness-of-fit test ($\chi^2 = 707.232$, $P = 0.943$) yielded non-significant results, indicating excellent model fit. Furthermore, multicollinearity diagnostics confirmed that the Variance Inflation Factors (VIF) for all independent variables were well below 5 (ranging from 1.053 to 3.881), ruling out significant multicollinearity. The variable assignments are shown in Table 2.

Variable coding was defined solely for regression modeling purposes. The assignment values do not imply biological directionality or clinical ranking and were used only to facilitate model estimation.

Independent risk factors for cervical lesions

Using the Inflammation group as the reference category, the multinomial logistic regression analysis evaluated the independent associations between the clinical variables and the risk of developing LSIL and HSIL (Table 3).

Risk factors for LSIL

Compared to the Inflammation group, both general high-risk HPV infection (OR = 3.930, 95% CI: 1.225–12.605, $P = 0.021$) and very-high-risk HPV infection (OR = 3.725, 95% CI: 1.137–12.206, $P = 0.030$) significantly increased the risk of LSIL. Furthermore, a high-risk TCT result was identified as an independent risk factor for LSIL (OR = 2.435, 95% CI: 1.408–4.212, $P = 0.001$). However, PAX1 and JAM3 methylation positivity were not significantly associated with LSIL ($P > 0.05$). Risk Factors for HSIL:

Table 2 Assignment of variables in the multinomial logistic regression model

Variable		Assignment
Marital status	X1	Unmarried = 0, Married = 1
Mode of delivery	X2	No delivery = 0, Vaginal delivery = 1, Cesarean section = 2
Postcoital bleeding	X3	No = 1, Yes = 0
Menopausal status	X4	Yes = 1, No = 0
HPV vaccination	X5	No = 1, Yes = 0
Contraceptive method	X6	No contraception = 0, Condom = 1, IUD = 2, Withdrawal = 3
Family history of cancer	X7	No = 1, Yes = 0
HPV infection	X8	Low-risk or negative = 3, Other high-risk = 2, Very-high-risk = 1
TCT	X9	NILM = 2, NonNILM = 1
P1	X10	Negative = 1, Positive = 0
J3	X11	Negative = 1, Positive = 0
Gravidity	X12	0 time = 2, 1–2 times = 1, ≥ 3 times = 0
Parity	X13	0 time = 2, 1–2 times = 1, ≥ 3 times = 0
Number of sexual partners	X14	0–1 = 2, ≥ 2 = 1
Pathology	Y	Inflammation = 1, LSIL = 2, HSIL = 3

IUD Intrauterine Device, *HPV* Human Papillomavirus, *TCT* ThinPrep Cytologic Test, *NILM* Negative for Intraepithelial Lesion or Malignancy, *Non-NILM* Abnormal TCT result, *LSIL* Low-Grade Squamous Intraepithelial Lesion, *HSIL* High-Grade Squamous Intraepithelial Lesion

Table 3 Multinomial logistic regression analysis of independent risk factors associated with cervical lesions

Variable	Category	LSIL-Inflammation				HSIL-Inflammation			
		OR	95% CI		*P*	OR	95% CI		*P*
			Upper Limit	Lower Limit			LowerLimit	Upper Limit	
Age		1.014	0.971	1.058	0.532	0.980	0.938	1.024	0.366
Gravidity									
	0	Reference				Reference			
	1–2	0.751	0.306	1.846	0.533	1.351	0.610	2.992	0.459
	≥ 3	1.495	0.571	3.916	0.413	1.624	0.648	4.075	0.301
Age at first sexual intercourse		0.995	0.9	1.102	0.930	0.925	0.832	1.028	0.148
Number of sexual partners									
	0–1	Reference				Reference			
	≥ 2	0.939	0.49	1.802	0.851	1.415	0.755	2.654	0.279
Menopausal status									
	Yes	Reference				Reference			
	No	1.700	0.649	4.453	0.281	1.548	0.546	4.389	0.411
HPV									
	Low-risk or negative	Reference				Reference			
	Other high-risk	3.930	1.225	12.605	0.021	2.930	0.854	10.058	0.088
	Very-high-risk (16, 18)	3.725	1.137	12.206	0.030	6.801	1.998	23.151	0.002
TCT									
	NILM	Reference				Reference			
	NonNILM	2.435	1.408	4.212	0.001	2.399	1.398	4.119	0.002
P1									
	Negative	Reference				Reference			
	Positive	1.953	0.539	7.082	0.308	4.145	1.310	13.119	0.016
J3									
	Negative	Reference				Reference			
	Positive	2.053	0.630	6.689	0.233	8.215	2.907	23.213	< 0.001

Abbreviations: *OR* Odds Ratio, *CI* Confidence Interval, *LSIL* Low-Grade Squamous Intraepithelial Lesion, *HSIL* High-Grade Squamous Intraepithelial Lesion, *HPV* Human Papillomavirus, *TCT* ThinPrep Cytologic Test, *NILM* Negative for Intraepithelial Lesion or Malignancy, *Non-NILM* Abnormal TCT result

The dependent variable for the multinomial logistic regression model is pathological diagnosis, with the "Inflammation" group acting as the reference category

The disease progression to HSIL was profoundly associated with specific viral, cytological, and epigenetic factors. Very-high-risk HPV infection significantly elevated the risk of HSIL compared to inflammation (OR = 6.801, 95% CI: 1.998–23.151, $P = 0.002$). Similarly, high-risk TCT results remained a strong risk factor for HSIL (OR = 2.399, 95% CI: 1.398–4.119, $P = 0.002$). Most notably, positive methylation results for both specific genes were robustly associated with high-grade lesions: patients with PAX1 positivity (OR = 4.145, 95% CI: 1.310–13.119, $P = 0.016$) and JAM3 positivity (OR = 8.215, 95% CI: 2.907–23.213, $P < 0.001$) had substantially higher odds of presenting with HSIL compared to the inflammation group.

Discussion

Cervical cancer is the fourth most common cancer among women globally, and its prevention heavily relies on the early identification and management of precancerous lesions [12]. Studies have shown that behavioral factors such as the number of sexual partners [13], an earlier age at first sexual intercourse [14], and gravidity [15] are positively correlated with the development of cervical cancer. Premenopausal women are generally more sexually active than postmenopausal women, and the probability of developing high-grade cervical lesions is higher in this group [16].Furthermore, research has indicated that the PAX1 and JAM3 genes play crucial roles in the occurrence and progression of cervical cancer [17]: approximately 30% of high-grade cervical lesions progress to cervical cancer [18–19].This study aimed to explore the association between specific clinical biomarkers—particularly PAX1 and JAM3 gene methylation—and the severity of cervical lesions in a high-risk colposcopy-referred cohort, identifying independent risk factors to aid in clinical risk stratification.

Our multinomial logistic regression analysis confirmed that both HR-HPV and abnormal TCT results are strongly associated with an increased risk of early cervical lesions. Specifically, compared to the inflammation group, both general high-risk HPV and very-high-risk HPV (such as 16/18) infections were significantly associated with Low-Grade Squamous Intraepithelial Lesions (LSIL), showing a similarly increased risk without a definitive dose-response trend [20]. Abnormal TCT results were independently associated with both LSIL and High-Grade Squamous Intraepithelial Lesions (HSIL). These findings align with established clinical knowledge that persistent HPV infection and cytologic abnormalities are primary drivers of early cervical pathogenesis [21].

However, the most notable finding of our study is the robust association between PAX1 and JAM3 gene methylation and the risk of HSIL. Our data demonstrated that PAX1 positivity (OR = 4.145) and JAM3 positivity (OR = 8.215) significantly increased the risk of HSIL compared to inflammation. Aberrant DNA methylation, particularly the hypermethylation of tumor suppressor gene promoters, is widely recognized as a critical early epigenetic event in carcinogenesis [22]. PAX1 (Paired box 1) and JAM3 (Junctional adhesion molecule 3) act as tumor suppressors, and their epigenetic silencing has been closely linked to the malignant transformation of cervical epithelial cells [17–23]. In a high-risk colposcopy-referred cohort, the strong association between the methylation of these specific genes and high-grade lesions suggests that they are powerful indicators of disease severity.Some estimated odds ratios were accompanied by relatively wide confidence intervals, which may reflect the single-center design, the enrichment of high-risk individuals within this cohort, and the limited sample sizes of certain subgroups. Therefore, these effect estimates should be interpreted cautiously and validated in larger multicenter studies.

Importantly, rather than proposing these markers as standalone primary screening tools, our findings highlight their potential value in risk stratification for patients who have already yielded abnormal initial screening results. In a population already referred for colposcopy, relying solely on HPV or TCT may not sufficiently differentiate those with true high-grade lesions from those with transient infections. The significant Odds Ratios observed for PAX1 and JAM3 methylation suggest that incorporating these epigenetic markers could help clinicians identify patients at the highest risk of HSIL, potentially prioritizing them for immediate intervention while avoiding overtreatment in lower-risk individuals [17] These findings suggest that PAX1/JAM3 methylation testing may serve as a valuable triage and risk-stratification tool following abnormal HPV or cytology results. Rather than replacing established first-line screening approaches, such as HR-HPV testing and cytology, methylation biomarkers may help identify patients at the highest risk of HSIL and optimize referral and management strategies.

In conclusion, HR-HPV infection, abnormal TCT results, and PAX1/JAM3 gene methylation were independently associated with cervical lesion severity in this high-risk colposcopy-referred cohort. PAX1 and JAM3 methylation showed strong associations with HSIL and may provide additional information for risk stratification. However, prospective multicenter studies are required to further validate their clinical utility and generalizability.

Limitations

This study has several limitations. First, the study population was limited to patients referred for colposcopy due to abnormal screening results or clinical symptoms, which introduces selection bias and restricts the

generalizability of our findings to the broader screening population.

Second, the analysis focused on evaluating multivariate Odds Ratios (ORs) rather than standard diagnostic performance metrics such as sensitivity, specificity, or Area Under the ROC Curve. As a result, comparisons of predictive accuracy across variables were not conducted.

Third, as a retrospective, single-center study, our results may be influenced by unmeasured confounders and decision-related bias, as treatment decisions were made based on individual clinical judgment. Although we adjusted for treatment variations, such bias could still affect the results.

Additionally, the lack of external validation with an independent cohort limits the applicability of our findings to other populations.

Furthermore, because of the retrospective cross-sectional design, the present study could not assess the longitudinal predictive value of PAX1/JAM3 methylation for lesion progression, persistence, or regression over time. Consequently, the prognostic utility of these biomarkers remains to be determined.

To strengthen the validity and generalizability of the findings, future multicenter prospective studies with larger sample sizes, longer follow-up periods, and external validation cohorts are warranted.

Acknowledgements
The authors thank AiMi Academic Services (www.aimieditor.com) for English language editing and review services.

Authors' contributions
Yahui Bian and Xiaodong Li proposed the research idea and supervised the whole process of the research and approved the final manuscript. Tongyao Geng conducted literature retrieval and data extraction.Sijia Li participated in the data analysis process.Yanan Bian drafted the All authors read.

Funding
This study was supported with the Medical Science Research Project of Hebei (No. 20240486).

Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the Ethics Committee of the First Hospital of Hebei Medical University (Ethics Number: V1.0, 20221107). Initially, approval was granted for a pilot cohort of 100 participants. To ensure adequate statistical power, additional approval was obtained to expand the cohort to 409 participants. Written informed consent was obtained from all participants at the time of enrolment, and all samples were collected in compliance with ethical guidelines. This study, involving human participants and identifiable data, was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Department of Gynecology, First Hospital of Hebei Medical University, Shijiazhuang, China
[2]Hebei Medical University, Shijiazhuang, China
[3]Department of Supplies, First Hospital of Hebei Medical University, Shijiazhuang, China

Received: 17 February 2026 / Accepted: 15 June 2026
Published online: 25 June 2026

References

1. Zhonghua SHANG, et al. Trends in Cervical Cancer Burden in China from 1990 to 2021 and Prediction for 2022 to 2035. Chin Gen Pract [J]. 2025;28(26):3314–20.
2. Yun ZHAO, et al. Screening for cervical cancer and diversion of abnormal screening results. J Chin Physician [J]. 2023;25(5):649–51. 655.).
3. Jing CAI, Zehua WANG. Challenges in screening and early detection of gynecologic malignancies. J Chin Physician [J]. 2023;25(5):641–3.
4. Su H, et al. The triage role of cytological DNA methylation in women with non-16/18, specifically genotyping high-risk HPV infection. Br J Cancer. 2025;132(11):1064–71.
5. Narkcham S, et al. Rescreening of high-risk HPV positive Papanicolaou tests initially screened as negative is a low yield procedure in the era of HPV genotyping. J Am Soc Cytopathol. 2021;10(6):558–64.
6. Koliopoulos G, et al. Cytology versus HPV testing for cervical cancer screening in the general population. Cochrane Database Syst Rev. 2017;8(8):CD008587.
7. Bowden SJ, et al. DNA methylation signatures of cervical pre-invasive and invasive disease: An epigenome-wide association study. Int J Cancer. 2025;157(2):305–16.
8. Zhang R, et al. PAX1/SOX1 gene methylation as a detection and triage method for triage of high risk HPV-Positive women in cervical cancer screening. Gynecol Oncol Rep. 2025;60(8):101794.
9. Wang H, et al. Epigenetic silencing of JAM3 promoted progression in serous ovarian carcinoma through PI3K/AKT pathway. Epigenomics. 2025;17(12):803–15.
10. Sun D, et al. The performance of JAM3/PAX1 methylation in the diagnosis of high-grade squamous intraepithelial lesions for women with high-risk HPV infection. BMC Cancer. 2024;24(1):1514.
11. Jing Fei L, Zhai,Jing, Wang, et al. Evaluating PAX1/JAM3 methylation for triage in HPV 16/18-infected women. Clin Epigenetics. 2024;16(1):190. https://doi.org/10.1186/s13148-024-01804-w.
12. Bray F, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2024;74(3):229–63.
13. Hu S, et al. Risk factors for hrHPV infection with cervical lesions patient in Anhui Province of China from 2021 to 2024: A retrospective study. Med (Baltim). 2025;104(30):e43384.
14. Mingrui SU, et al. Characteristics of vaginal microbiota in various cervical lesions in women infected with HR-HPV species alpha-9. Chin J Front Med Science(Electronic Version) [J]. 2024;16(1):26–35.
15. Sun L, et al. Optimization of Cervical Cancer Screening: A Stacking-Integrated Machine Learning Algorithm Based on Demographic, Behavioral, and Clinical Factors. Front Oncol. 2022;12(2):821453.
16. Junting HUANG, et al. A clinical study on screening and management of high-risk HPV infections and cervical lesions in postmenopausal women. Chin J Practical Gynecol Obstet [J]. 2025;41(5):548–52.
17. Li X, et al. High-grade cervical lesions diagnosed by JAM3/PAX1 methylation in high-risk human papillomavirus-infected patients. Zhong Nan Da Xue Xue Bao Yi Xue Ban. 2023;48(12):1820–9.
18. Xinxin CHEN, Zhifang CHEN. Research progress of cervical cancer and precancerous lesions screening. China Med [J]. 2025;20(4):636–40.
19. Sustova P, et al. The Predictive Impact of HPV Genotypes, Tumor Suppressors and Local Immune Response in the Regression of Cervical Intraepithelial Neoplasia 2–3: A Prospective Population-Based Cohort Study. Int J Mol Sci. 2025;26(11):5205. https://doi.org/10.3390/ijms26115205.

20. Ding H, Jin H, Tang Y. HPV genotype distribution and cervical lesions in Chongqing: a comprehensive analysis of 229,770 females (2015–2023). BMC Infect Dis. 2025;25(1):760.
21. Zhou J, et al. The Value of P16 Protein Detection in the Screening for High-Grade Squamous Intraepithelial and Higher Lesions of the Cervix by the Combined Detection of HR-HPV and TCT. Diagn Cytopathol. 2025;53(6):283–9.
22. Shang X, et al. [A multicenter study on the accuracy of PAX1/JAM3 dual genes methylation testing for screening cervical cancer]. Zhonghua Yi Xue Za Zhi. 2024;104(20):1852–9.
23. Fei J, et al. Evaluating PAX1/JAM3 methylation for triage in HPV 16/18-infected women. Clin Epigenetics. 2024;16(1):190.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.